

Course content and style

International Marketing Analytics



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The journey begins



Instructor: Dr Arne Floh

- **Position:** Senior Lecturer in International Marketing, Institute for International Business, WU Vienna
- **Also:** Deputy Programme Director, Export and Internationalisation Management
- **Teaching:** International Marketing, Marketing Analytics, Digital Marketing and Social Media, Entrepreneurial Marketing
- **Research:** Consumer behaviour, relationship marketing, electronic marketing and social media
- **Contact:** arne.floh@wu.ac.at, office hours by appointment (see closing slide)



Instructor: background

- **Experience:** international and industry background, more than 20 years of teaching
- **Supervision:** BSc, MSc and PhD theses
- **Private:** football, electronic gadgets, Austrian wine



- **Marketing analytics** turns marketing data into decisions: which markets to invest in, how to price, how much to spend on which channel, and whether it worked.
- **One question dominates** international marketing budgets today: how should a fixed media budget be split across countries and channels?
- **Marketing mix modelling (MMM)** is back at the centre of this debate since privacy changes made user-level tracking unreliable. Managers who can read, challenge and commission these models hold the budget conversation.
- **One running case:** Alpenglöck, a Vienna-based premium chocolate brand active in six European markets (AT, DE, FR, IT, NL, PL).
- **Python with AI assistants:** the assistant does most of the typing; you learn to specify, run, check and communicate analyses.

Five sessions, five decisions

- Each session starts from a decision an international marketing manager has to take and works backwards to the model that answers it.
- **Session 1:** Is our price too high in Poland? Price and promotion elasticities across countries
- **Session 2:** What did TV do for us in Germany? Carryover, saturation and return on ad spend
- **Session 3:** Where should the next euro go? Budget allocation across six countries and five channels
- **Session 4:** Can we trust the model? Sales forecasts, geo-lift experiments and multi-market attribution
- **Session 5:** Board meeting: the groups pitch and defend their 2027 budget

Learning outcomes

On successful completion of this module, students will be able to:

- 1. Explain marketing mix modelling:** carryover (adstock), saturation, baseline versus incremental sales, ROAS versus marginal ROAS.
- 2. Estimate market response models:** specify, estimate and diagnose regression models in Python, from OLS and log-log elasticities to panel fixed effects and hierarchical Bayesian MMM.
- 3. Allocate budgets across markets:** translate model output into a cross-country, cross-channel allocation and defend it under uncertainty.
- 4. Forecast sales:** produce and evaluate short-term forecasts as the baseline for planning.
- 5. Design experiments:** design and analyse a geo-lift test and use it to validate and calibrate an MMM.
- 6. Compare attribution approaches:** explain why last-touch, regression-based and Shapley attribution disagree across markets.

Learning outcomes (continued)

- 7. Work reproducibly:** use Python, Positron, Quarto and GitHub to build analyses that colleagues can rerun and audit.
- 8. Use AI assistants responsibly:** direct an AI coding assistant, verify its output and document the prompts behind an analysis.
- 9. Communicate:** present analytical findings to a board audience in a written report and a pitch.
- 10. Work in teams:** collaborate in version-controlled analytical work and take collective data-driven decisions.
- 11. Think critically:** evaluate data sources, model assumptions and findings in the context of international marketing.

Methods of teaching and learning

In the classroom

- Five in-person sessions on Tuesdays, 6 October to 3 November 2026
- Concept lecture (60 min), guided lab in Positron (90 min), team exercise on the project data (60 min), debrief
- Coding is AI-assisted: your laptop, your assistant, your checks
- Everything lives in GitHub: pull, commit and push from Session 1

Between sessions

- Read the book chapters and session materials in advance
- Online quiz after each of Sessions 1 to 4
- Lab check-ins after Sessions 2 and 4
- Group project work in your team repository; consulting hours on request
- Q&A forum on Canvas

Module roadmap

Session	Date	Topic	Methods	Chapter
1	Tue 6 Oct	Course content and style, toolkit, regression foundations	Descriptive statistics, OLS, log-log elasticities, dummies and interactions	1, 11
2	Tue 13 Oct	Marketing mix modelling I: one country	Adstock, saturation, decomposition, ROAS, diagnostics	3
3	Tue 20 Oct	Marketing mix modelling II: many countries, one budget	Panel fixed effects, hierarchical Bayesian MMM, budget optimisation	3
4	Tue 27 Oct	Forecasting, experiments and attribution	Time series regression, difference-in-differences, logistic regression, Shapley values	9, 4
5	Tue 3 Nov	Group project pitches and wrap-up	Board presentation, ground-truth reveal	11

Components

- **5 %** Online certificates Python and GitHub (individual)
- **5 %** Self-reflection (individual)
- **20 %** Online quizzes, 4 × 5 % (individual)
- **20 %** Lab check-ins, 2 × 10 % (individual)
- **40 %** Group project: report, repository and pitch
- **10 %** Peer review

Rules

- **Groups:** 5 students, enrol on Canvas by the end of Session 1
- **Grades:** WU scale, 1 from 90 %, 2 from 80 %, 3 from 70 %, 4 from 60 %
- **Attendance:** at least 80 % of sessions
- **Late submissions:** minus 10 percentage points per day unless agreed in advance
- **AI tools:** expected for coding; you remain responsible for every number you report

Online certificates

- Complete two free online courses before the second session:
 - **Kaggle Learn: Python** (about 5 hours, certificate on completion)
 - **GitHub Skills: Introduction to GitHub** (about 1 hour)
- Install the course environment with the setup guide: Python via uv, Positron, Quarto, GitHub Desktop, and run the installation check
- **Upload** both certificates and a screenshot of the passing installation check on Canvas by Mon 12 Oct, 11:59 pm

Quizzes and lab check-ins

Online quizzes (individual)

- After each of Sessions 1 to 4
- Open Wednesday 9 am, close Monday 11:59 pm
- 10 multiple-choice questions in 10 minutes
- Closed book: concepts, not code
- 5 % each

Lab check-ins (individual)

- After Sessions 2 and 4
- Replicate the lab for another country or dataset with your AI assistant
- Submit the Quarto notebook, rendered HTML, result files and a short prompt log on Canvas
- Auto-checked for plausibility, reviewed for interpretation
- 10 % each

Group project

- **The task:** as Alpenglow's analytics team, recommend to the board how the 2027 media budget should be split across six countries and five channels.
- **Deliverables:** GitHub repository with a Quarto report (max 12 pages), allocation_2027.csv, a pitch deck (max 8 slides) and a prompt log.
- **Milestones:** repository and data exploration before Session 2; first model table before Session 4.
- **Support:** online consulting hours (registration needed).
- **Deadline:** Mon 2 Nov 2026, 6 pm. Pitches in Session 5 on Tue 3 Nov 2026.

- Every group member rates the other members on a 0 to 10 scale
- Six criteria: tasks carried out, deadlines met, quality of work, communication and respect, attendance, pre-agreed rules
- Kept in strict confidence; feeds the peer review component (10 %)
- Submit on Canvas with the final project

Peer-Rating-Form

Each group member is required to submit a peer rating form. This form must contain an evaluation (i.e. a grade) of each group member. Do not assign yourself a grade as this would defeat the idea of peer rating. **The form will be kept in strict confidence.**

In the spaces provided below, please fill in your name, choose your group, and write complete names of all of your group members. Finally, assign a grade on a scale from 0 to 10 to each of your group members (**10 being the highest** and 0 being the lowest grade). Please assign grades based on the following criteria:

- → Willingness to carry out assigned tasks
- → Ability to meet deadlines
- → Quality of the individual's work
- → Communication ability and respectful treatment of others
- → Availability for and attendance at group meetings
- → Following the pre-agreed rules

Your Name: → Please enter your name

Your Group: → Choose your group

Group Member Names	Grade
1.Group Member Name	Insert grade
2.Group Member Name	Insert grade
3.Group Member Name	Insert grade
4.Group Member Name	Insert grade
5.Group Member Name	Insert grade



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